SIC-MMAB: Synchronisation involves communication

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Overview

Multiplayer bandits problem

SIC-MMAB

Contradiction with lower bounds

Dynamic setting

Related works

Multiplayer bandits problem

Introduction

Motivation: **Cognitive Radio** (5G)

- Optimize spectrum access for **Primary** and Secondary users
- when **Primary** user on channel $k \rightarrow$ priority over Secondary users
- when several Secondary on same channel: interference/collision

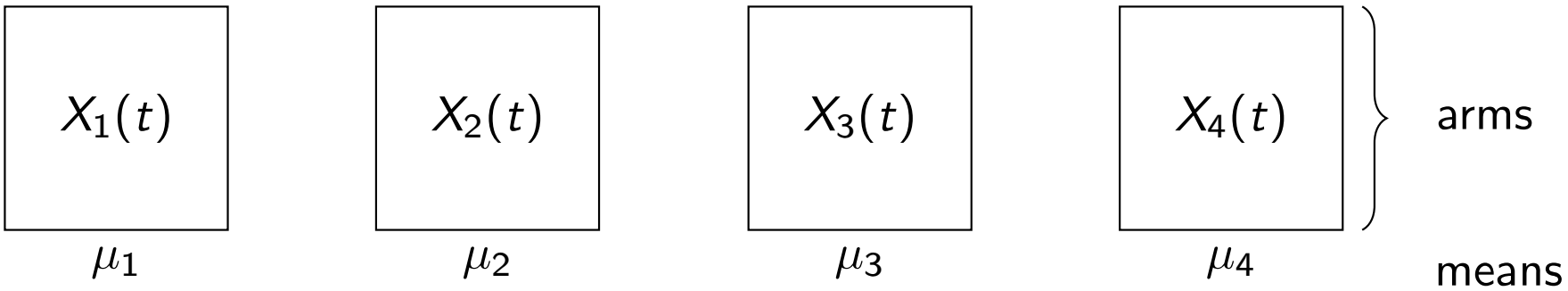
Goal for secondary users: find and communicate on best channels

Bandit game at round $t \in \{1, \dots, T\}$

K arms

Player

i.i.d. $X_k(t) \sim \mathcal{B}(\mu_k)$ in $[0, 1]$
pull arm $\pi(t)$ given past
observe reward $X_{\pi(t)}(t)$



$$X_k(t) = \begin{cases} 0 \text{ if } \textbf{Primary} \text{ user on } k \\ 1 \text{ otherwise} \end{cases}$$

Bandit game at round $t \in \{1, \dots, T\}$

K arms

Player

Pull arm 2

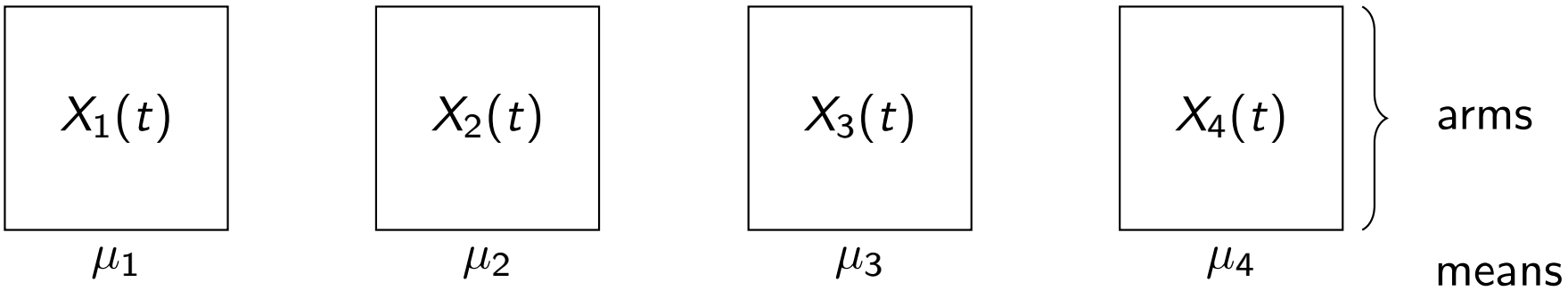
i.i.d. $X_k(t) \sim \mathcal{B}(\mu_k)$ in $[0, 1]$
pull arm $\pi(t)$ given past
observe reward $X_{\pi(t)}(t)$

$X_1(t)$	$\mathbf{X_2(t)}$	$X_3(t)$	$X_4(t)$	arms
μ_1	μ_2	μ_3	μ_4	means

$$X_k(t) = \begin{cases} 0 \text{ if } \textbf{Primary} \text{ user on } k \\ 1 \text{ otherwise} \end{cases}$$

Multiplayer Bandit game at round $t \in \{1, \ldots, T\}$
K arms, M players

Player 1 Player 2 Player 3



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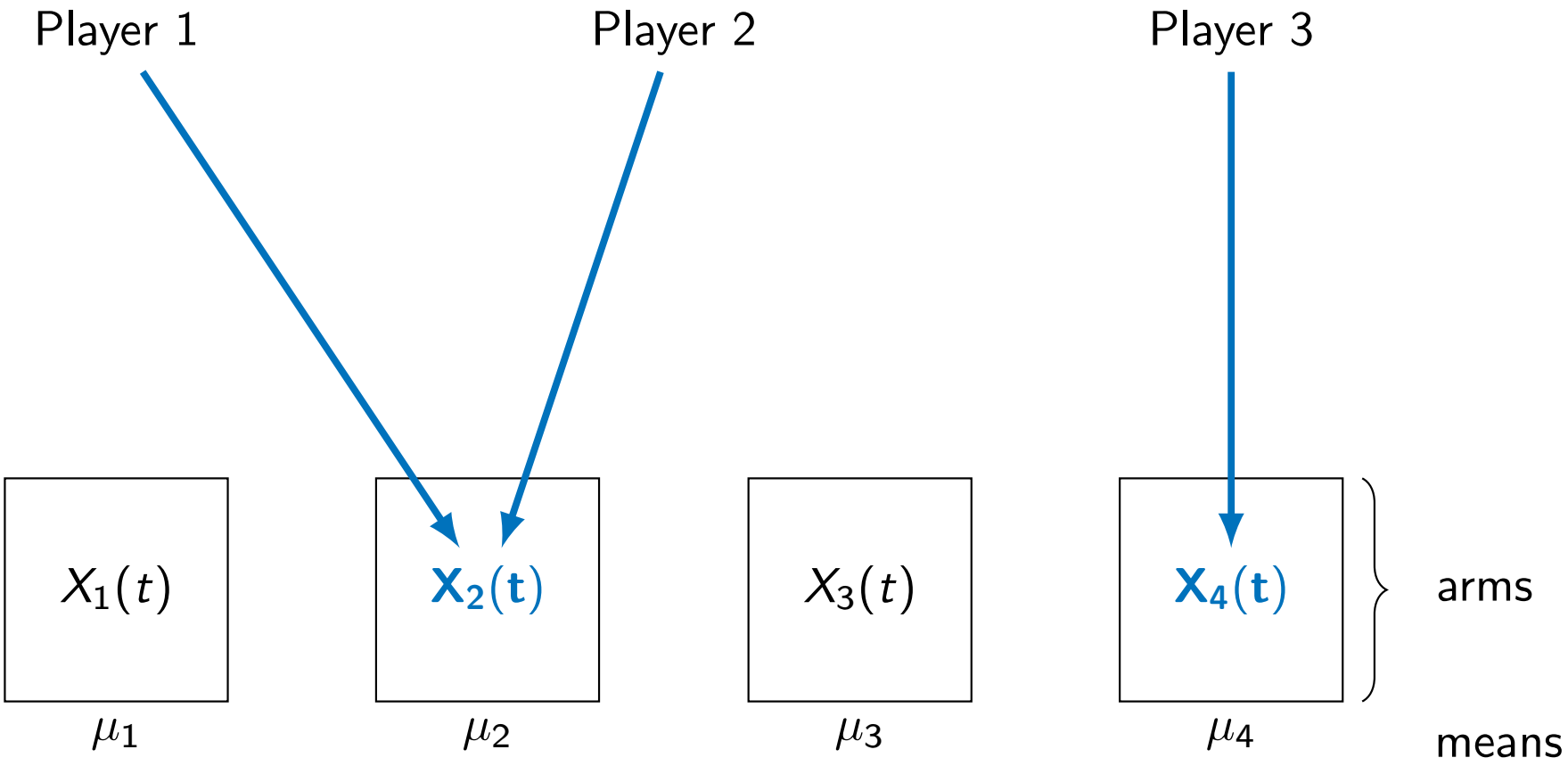

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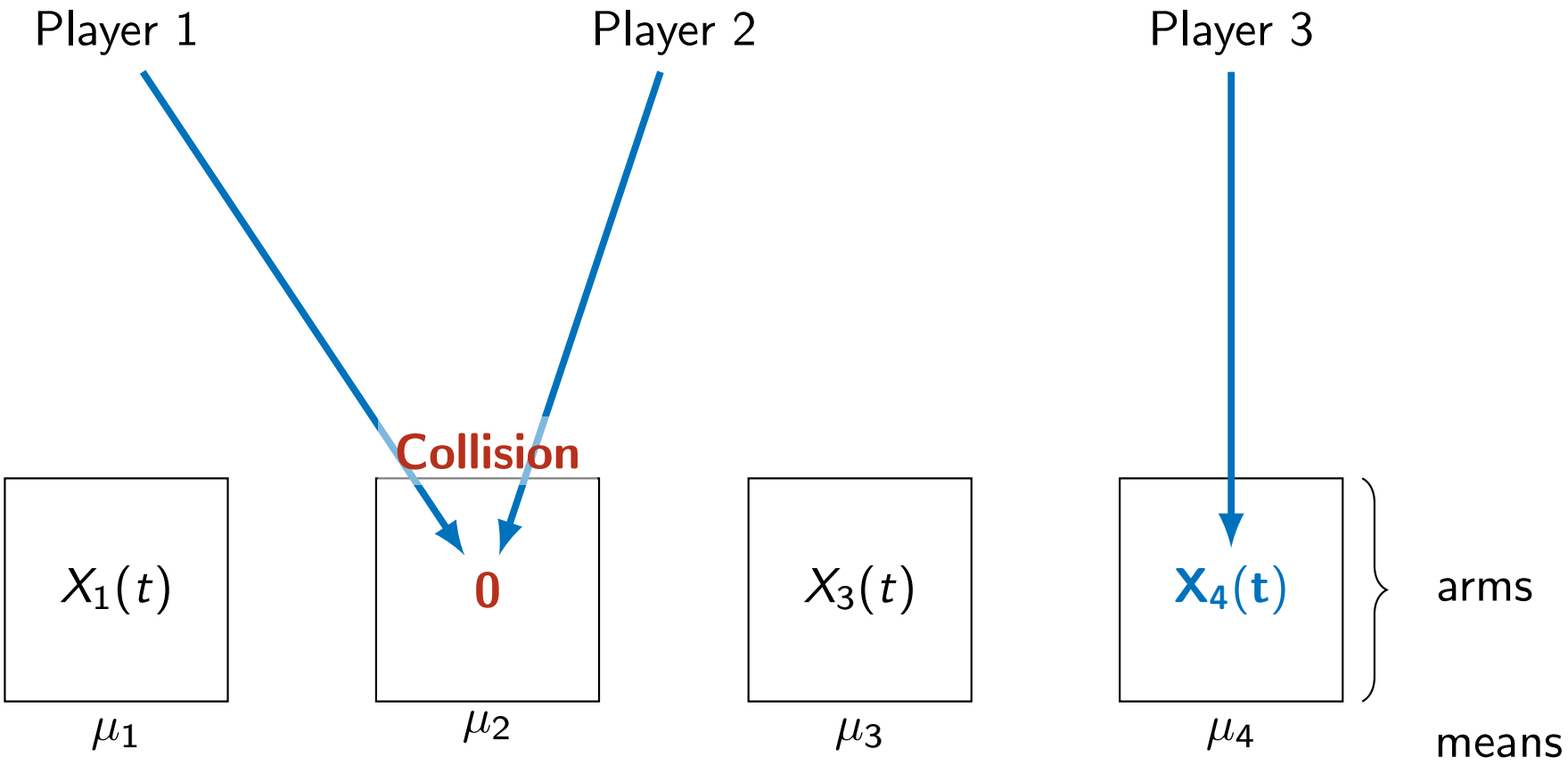

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Model: Multiplayer Multi-Armed Bandits

- K arms with Bernoulli rewards $X_k(t) \sim \mathcal{B}(\mu_k)$

$$\text{w.l.o.g. } \mu_1 \geq \mu_2 \geq \ldots \geq \mu_K$$

- $M \leq K$ players pull arms $\pi^j(t)$ simultaneously for $t = 1, \ldots, T$
- **Decentralized**: players can not communicate & M is unknown
- get reward $r^j(t) = X_{\pi^j(t)}(t) \mathbb{1}_{\text{no collision on } \pi^j(t)}$

$$\textbf{Regret: } R_T = T \sum_{k=1}^{M} \mu_k - \mathbb{E}_\mu \left[\sum_{t=1}^{T} \sum_{j=1}^{M} r^j(t) \right]$$

Feedback/sensing settings

$$r^j(t) = X_{\pi^j(t)}(t)\mathbb{1}_{\text{no collision on } \pi^j(t)}$$

- Collision sensing: observe $r^j(t)$ and $\mathbb{1}_{\text{no collision on } \pi^j(t)}$
- No sensing: observe only $r^j(t)$
- Statistic sensing: observe $r^j(t)$ and $X_{\pi^j(t)}(t)$

Collision Sensing: SIC-MMAB

Centralized case

- Players communicate (for free) $\rightarrow$ **no** collision
- Combinatorial bandits, tight bound:
  [Anantharam et al., 1987, Komiyama et al., 2015]

Regret in $\sum_{k>M} \frac{\log(T)}{\mu_k - \mu_M}$

Lower bounds

Centralized lower bound

$$\sum_{k>M} \frac{\log(T)}{\mu_M - \mu_k}$$

[Anantharam et al., 1987]

Decentralized lower bound

$$M \sum_{k>M} \frac{\log(T)}{\mu_M - \mu_k}$$

[Liu and Zhao, 2010]
[Besson and Kaufmann, 2018]

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SIC-MMAB

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SIC-MMAB

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Decentralized $\sim$ Centralized

How is this possible?

Main trick

Observation: $\mathbb{1}_{\text{no collision on } k} \in \{0, 1\}$ seen as a bit sent between players

- force collisions during communication rounds
- when i *talks* to j: $\begin{cases} \text{collide with } j \text{ to send a 1 bit} \\ \text{do not collide to send a 0} \end{cases}$
- players communicate empirical means to each other
  $\rightarrow$ centralization
- sublogarithmic number of communication rounds

Algorithm structure

Algorithm 1: SIC-MMAB

Initialization Phase

for $p = 1, ..., \infty$ **do**

  Exploration phase p for 2^p rounds

  Communication phase p

  Accept/reject (sub)-optimal arms

end

Exploitation phase: pull optimal arms until T

Initialization phase

- Orthogonalize players: **Musical Chairs** for $K \log(T)$ rounds [Rosenski et al., 2016]
  - Sample arm k uniformly at random
  - If collision $\rightarrow$ continue
  - No collision $\rightarrow$ stick to arm k until $K \log(T)$

With proba $1 - M/T$, all players end on different arms

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- Compute M and rank j: **Sequential Hopping**
  - player on arm k waits for $2k$ rounds
  - player then *hops* for $2(K - k)$ rounds
  - $M - 1 =$ number of collisions and
    $j - 1 =$ number of collisions for the $2k$ first rounds

Exploration phase p

- each player explores each arm 2^p rounds
  - start at different positions given by ranks
  - sequential hopping $\rightarrow$ **no collision**
- player j gathered statistics on arm k
  - $S_k^j(p)$ rewards 1
  - $T_k^j(p)$ pulls

Communication phase p

- player i communicates $S_k^i(p) \in [2^p]$ to player j:
  - encoded in p bits $(0, 1, 0, \ldots, 0)$
  - send it in p rounds: (no coll., coll., no coll., ..., no coll.)
- players communicate **one at a time**
- they know when and how to do so, thanks to their ranks j
- possible quantization for non binary rewards

length of comm. phase p: KM^2p

Algorithm structure

Algorithm 2: SIC-MMAB

Initialization Phase
for $p = 1, ..., \infty$ **do**
  Exploration phase p
  Communication phase p
  Accept/reject (sub)-optimal arms
end
Exploitation phase

Accept/Eliminate (sub)-optimal arms

All players have the same **centralized** empirical means $\hat{\mu}_k$

Concentration inequality (Hoeffding)

With high proba, $|\mu_k - \hat{\mu}_k| \leq \sqrt{2\log(T)/T_k(p)}$

$\rightarrow$ arm k is detected better than l if:

$$\hat{\mu}_k - \sqrt{2\log(T)/T_k(p)} \geq \hat{\mu}_l + \sqrt{2\log(T)/T_l(p)}$$

happens after $\frac{\log(T)}{(\mu_k - \mu_l)^2}$ pulls

Accept/Eliminate (sub)-optimal arms

- arm k sub-optimal if M arms are detected better
  $\rightarrow$ eliminated from the set to explore
- arm k optimal if $K - M$ arms are detected worse
  $\rightarrow$ attributed to player with largest rank

$\rightarrow$ exploration ends after $N = \log\left(\frac{\log(T)}{(\mu_M - \mu_{M+1})^2})\right)$ phases

Regret bound

Initialization: $M \times \text{length} \simeq MK\log(T)$

Communication: $M \times \sum_{p=1}^{N} pM^2K \simeq M^3K\log^2(\frac{\log(T)}{(\mu_M-\mu_{M+1})^2})$

Exploration: centralized regret bound $\sum_{k>M} \frac{\log(T)}{\mu_M-\mu_k}$

Low probability events: $o(\log(T))$

Total regret

$$R_T \lesssim \sum_{k>M} \frac{\log(T)}{\mu_M - \mu_k} + MK\log(T)$$

Contradiction with lower bounds

Contradict the lower bound?

Recall

Lower bound $M \sum_{k>M} \frac{\log(T)}{\mu_k - \mu_M}$

SIC-MMAB $\sum_{k>M} \frac{\log(T)}{\mu_k - \mu_M} + KM \log(T)$

Why this contradiction?

- Lower bound proofs assumed that best algorithms do not collide
- Wrong: SIC-MMAB deduces a lot of information from collisions
- Decentralized **as hard as** centralized

Towards a better model?

- SIC-MMAB uses unrealistic/undesired communication protocols
- abuses from a loophole allowing it
- need for a better model, without such a loophole
- which model assumption did go wrong?

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  - collision sensing?

No sensing setting

Assumption

Known lower bounds $\mu_k \geq \mu_{\min} > 0$

Observation: we can send a bit with high proba. in $\log(T)/\mu_{\min}$ rounds

Algo 1 SIC-MMAB with $\log(T)/\mu_{\min}$ comm. rounds instead of 1

- comm. regret becomes $M^3 K \frac{\log(T)}{\mu_{\min}} \log^2(\log(T))$

Algo 2 limited & different communication

- do not communicate statistics but only when an arm is found (sub)-optimal
- regret in $M \sum_{k>M} \frac{\log(T)}{\mu_k - \mu_M} + \frac{MK^2}{\mu_{\min}} \log(T)$

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  - cooperative players? (work in progress)

Towards a better model?

- SIC-MMAB uses unrealistic/undesired communication protocols
- abuses from a **loophole** allowing such protocols
- which model assumption did go wrong?
  - collision sensing?
  - cooperative players? (work in progress)
  - synchronisation between players?
    $\rightarrow$ more realistic dynamic model

Dynamic setting: DYN-MMAB

Dynamic Model

Asynchronicity assumption

Player j enters game at unknown time $\tau^j \in [T]$ and stays until T.

- varying & unknown set of players $\mathcal{M}(t)$
- no synchronisation $\implies$ similar protocols are not possible
- No Sensing setting

A dynamic algorithm

Only 2 different states:

Exploration: sample arm **uniformly at random**

Exploitation: **occupy** some optimal arm until T

Three difficulties:

1. Detect arms occupied by other players
2. Estimate the best **available** arm
3. Start occupying the best available arm

Detect occupied arms

- If k occupied, rewards only 0
- If k not occupied, positive reward with proba $\mu_k(1-\frac{1}{K})^{M_t-1} \geq \frac{\mu_k}{e}$

For an occupied arm k

- if μ_k tightly estimated: after $\simeq \frac{e\log(T)}{\mu_k}$ successive 0, k is assumed occupied
- otherwise, $\hat{\mu}_k$ will quickly drop to 0 and k will become sub-optimal

Estimate **available** arms

Players sample uniformly at random $\implies \mathbb{E}[r_k(t)] = \mu_k(1 - \frac{1}{K})^{M_t - 1}$

Player estimates $\gamma_t \mu_k$ where $\gamma_t = \frac{1}{t} \sum_{s=\tau^j+1}^{\tau^j+t} (1 - \frac{1}{K})^{M_s}$

- $\mu_k \geq \mu_l \iff \gamma_t \mu_k \geq \gamma_t \mu_l$
- concentration inequalities for $\gamma_t \mu_k$ (when k still free)
- $\gamma_t \geq \frac{1}{e} \implies$ estimating $\gamma_t \mu_k$ instead of μ_k takes roughly same time

Player detects best available arm k after time $\mathcal{O}\left(\frac{K \log(T)}{(\mu_k - \mu_{k+1})^2}\right)$

Occupy best available arm

- Once arm detected as best available $\rightarrow$ try to occupy it
- **Continue** sampling uniformly at random
- positive reward $\rightarrow$ occupy that arm
- observe only 0 rewards ?
  - detect it as occupied
  - continue exploration until next available arm

At some point, succeed in occupying an arm, while all better arms occupied

Regret bound

New regret definition:

$$\sum_{t=1}^{T} \sum_{k=1}^{\mathrm{card}(\mathcal{M}(t))} \mu_k - \mathbb{E}_\mu \left[\sum_{t=1}^{T} \sum_{j \in \mathcal{M}(t)} r^j(t) \right]$$

Dynamic regret bound

$$R_T \lesssim \overbrace{\frac{MK \log(T)}{\bar{\Delta}_M^2}}^{\text{detection of optimal arms}} + \overbrace{\frac{M^2 K \log(T)}{\mu_M}}^{\text{detection of occupied arms}}$$

with $\bar{\Delta}_M = \min_{k \leq M} \mu_k - \mu_{k+1}$

Drawback: quadratic dependence in Δ (due to uniform sampling)

Some related works (in random order)

Adversarial case

[Bubeck et al., 2019] considered adversarial rewards $X_k(t)$

- $\sqrt{T}$ regret for 2 players
- uses communication trick to coordinate players:
  - one with high frequency switches
  - the other with low frequency switches

Improving SIC-MMAB

Heterogeneous case: [Boursier et al., 2019]

- Arm means μ_k^j differ between players
- Improvement of comm. protocol: a leader gathers the information and decides for the others
- Do not eliminate arms, but player-arm pairs (j, k)

Improving SIC-MMAB

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- Do not eliminate arms, but player-arm pairs (j, k)

Optimal algorithm for homogeneous: [Proutiere and Wang, 2019]

- initialization in constant time (in T)
- exploration only by the leader
- regret $\leq \sum_{k>M} \frac{\log(T)}{\mu_M - \mu_k} + o(\log(T))$
- Confirms: decentralized is as hard as centralized

Other recent works

- Heterogeneous case:
  - similar protocols [Tibrewal et al., 2019]
  - implicit comm. through Markov chains [Bistritz and Leshem, 2018]
  - arms have preferences over players [Liu et al., 2019]
- No sensing [Lugosi and Mehrabian, 2018]
- Collision only implies drop in reward [Magesh and Veeravalli, 2019]

Recap & Open questions

Recap:

- Synchronisation allows communication protocols
- contradicts previous lower bounds: decentralized $\sim$ centralized
- synchronisation is a loophole in the model and has to be removed
- more realistic dynamic model: first logarithmic regret algorithm

Open questions:

- is the dynamic setting a *perfect* choice?
- room for improvement in hard settings (statistic sensing, adversarial rewards, heterogeneous, dynamic, etc.)

Thank you!

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